

Compensation of the Fluctuations of Differential Delay for Frequency Transfer in DWDM Networks

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Abstract—This paper investigates the possibility of improving the stability of radio-frequency transfer in telecommunication dense wavelength division multiplexing fiber-optic networks. As it has been identified, the dispersion compensation fibers (DCF), frequently used in these networks, cause substantial differential delay, whose temperature-induced fluctuations have the most significant impact on the deterioration of the stability of the frequency transfer. The authors present a method that allows achieving significant improvement in the long-term stability of the frequency transfer. The developed method is based on modeling the impact of DCFs with the help of remotely accessible temperature sensors factory installed by the manufactures in DCF modules. The effectiveness of the proposed solution has been tested on three different long-haul routes (up to 1550 km), set up in the operational Polish National Research and Education Network.

Index Terms—Alien wavelength, DWDM network, frequency transfer, optical fiber.

I. INTRODUCTION

THE problem of radio-frequency transfer in fiber optic networks has been widely investigated by researchers in recent years. Recently, the best results of the frequency transfer are obtained using a single fiber [1]–[7]. However, the lease of dark fibers is often not economically justified, or even impossible. For these reasons, many efforts are also being undertaken to develop methods for sharing fibers with currently existing telecommunications systems. Some of them, such as Precision Time Protocol or Synchronous Ethernet [8], are based on encapsulation of time stamps into a sequence of telecommunications data. Another technique is to use a narrow slice of the optical spectrum in a dense wavelength-division-multiplexing (DWDM) telecom system for transmitting nontelecom signals generated by terminals of a particular frequency transfer system in a so-called alien wavelength DWDM service [9]–[12].

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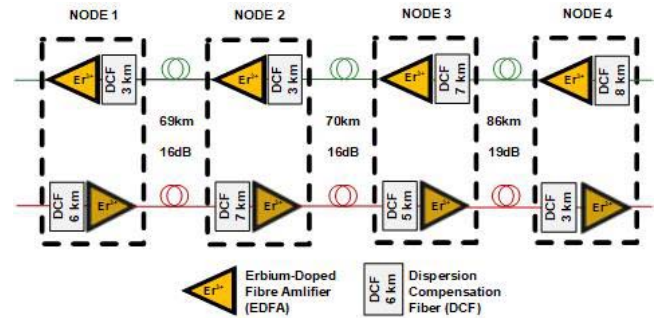


Fig. 1. Example of the arrangement (length and location) of DCF modules along an excerpt of the real DWDM line.

The fundamental challenge in the distribution of frequency using optical fibers is the need for compensation of the propagation delay/phase fluctuations, caused by environmental factors. A standard solution is to arrange a feedback system with forward and backward signals being exchanged between the terminal devices. In the case of bidirectional transmission in a single fiber, the very strong correlation (symmetry) of the fluctuations observed in the forward and backward directions allows obtaining efficient delay stabilization [7], [13]–[17]. However, in DWDM networks, the forward and backward transmissions are realized in parallel, but physically not in the same optical paths (unidirectional transmission) [4], [11], [18]–[20], which to some extent degrade the symmetry between the two directions, and thus deteriorate the delay stabilization.

There are two main sources of asymmetry between the forward and backward directions in DWDM networks: the optical cables connecting the network nodes and the nodes themselves. The environmental temperature fluctuations may affect two fibers in not perfectly the same way, even when the fibers are located in the same cable. Moreover, the so-called acoustic noise (related to local tensions, vibrations, and frictions) is not perfectly matched for two fibers [21]–[23]. The network nodes perform various processing of optical signals, such as amplification, optical routing, filtering, and dispersion compensation. As it was shown in [12] and [20], the source of a significant part of the differential (and thus uncompensated) frequency signal's delay fluctuations is related to the dispersion compensation fiber (DCF) modules, arranged in the form of fiber spools. It is a common situation that the lengths of these fibers are different for different directions in a given node (see Fig. 1).

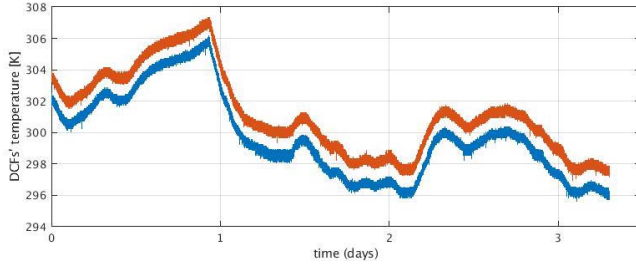


Fig. 2. Example of temperature fluctuations in two DCF modules, placed in adjacent slots of the equipment rack.

This paper presents the method of the correction of residual delay/phase changes by performing data postprocessing, based on the knowledge of the length of each individual DCF module and its temperature variations.

II. IMPACT OF DISPERSION COMPENSATORS ON FREQUENCY TRANSFER

DCFs in the form of spools with a special negative-dispersion fiber are widely used in DWDM systems. The lengths of DCF spools located in particular nodes are deliberately chosen during the network designing process, which, in practical cases, usually leads to a substantial difference in the lengths of the forward and backward direction DCFs at each node (see Fig. 1). In this situation, any change of the node temperature affects the forward and backward propagations delay differently, destroying the forward-backward symmetry, and thus degrades the stability of frequency transfer.

Knowing the thermal coefficient of the fiber propagation delay A , usually reported to be close to $38 \text{ ps}/(\text{km} \cdot \text{K})$ [24], [25], and taking into account that a half of the forward-backward propagation asymmetry is visible at the output of the delay stabilizing system [26], one could calculate the impact of DCFs' temperature on the differential delay fluctuations $\Delta\tau(t)$ using the following equation:

$$\Delta\tau(t) = \frac{1}{2}A \sum_{i=1}^N (L_{\text{DCF}_i} \cdot \Delta T_{\text{DCF}_i}(t) - L'_{\text{DCF}_i} \cdot \Delta T'_{\text{DCF}_i}(t)) \quad (1)$$

where L_{DCF_i} is the DCF length in kilometers, ΔT_{DCF_i} is the DCF temperature change and i runs through all N network nodes. The contributions from all DCFs were represented as “+” for the forward DCFs, and as “−” for the backward ones (marked with a prime).

The temperature fluctuations for both DCFs at a particular node are very similar (see Fig. 2), as they are usually located in neighboring slots of the same equipment rack, but unequal lengths of the spools are crucial. According to practical observations, temperature changes in typical telecommunication nodes range from a few to about dozen kelvin, and in the case of air-conditioning malfunction, even up to several tens of kelvin. Thus, the fluctuations of $\Delta\tau$ may be substantial, up to several nanoseconds.

Because most of the current systems have DCFs fitted with a factory-equipped temperature sensor, the temperatures

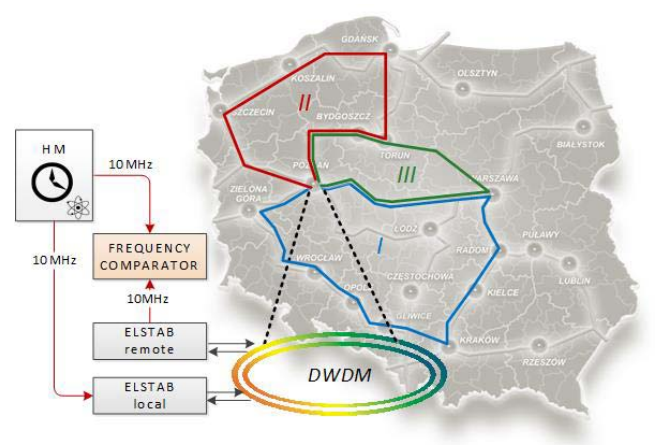


Fig. 3. Diagram of the experimental frequency transfer setup. Blue (I), red (II), and green (III) lines show geographical localizations of three different routes in DWDM network.

are easily accessible remotely via a network management system. This inspired us to exploit these data and propose the postprocessing method allowing to improve the stability of the frequency transfer.

III. EXPERIMENTAL SETUP

All experiments were arranged within the Polish National Research and Education Network (PIONIER network), where three different routes forming the geographical loops were set up (Fig. 3). In all three routes, the frequency signal was carried along with a number of 10-Gbps intensity-modulated signals transmitted in the same pair of fibers. The routes have the following lengths: the first (I) 1550 km, the second (II) 1380 km, and the last one (III) 1000 km. All of them were based on soil-deployed fibers. The network uses advanced reconfigurable add-drop multiplexers for optical wavelength routing, and all network equipment components were FSP3000 series devices made by ADVA Optical Networking. The lengths of individual DCF modules ranged from 1 to 10 km, with 1-km gradation. The temperature data from all DCF modules were collected using simple network management protocol (SNMP).

The frequency signal (10 MHz) from a passive hydrogen maser was connected to the AGH-developed electronically stabilized (ELSTAB) [26] terminals, which fed their optical signals directly into the DWDM system. Both terminals of ELSTAB (local and remote) were located at the same laboratory in Poznan. Block diagram of the setup exploited in the experiments is shown in Fig. 3.

The ELSTAB units, originally dedicated for bidirectional transmission in a single fiber, were adapted to the unidirectional transmission over the pair of “parallel” fibers in a single DWDM channel. For this purpose, both the local and remote lasers were tuned to the same DWDM channel, and also forward and backward optical signals were separated at different optical connectors. The basic idea of phase stabilization was, however, unchanged: phase fluctuations were sensed and canceled with the help of the backward signal coming back from the remote terminal, using a pair of electronic delay

lines, located in the forward and backward signal paths [26]. The stability of the frequency transfer was assessed by measuring the residual phase fluctuations between the input reference signal (10-MHz frequency supplied to the ELSTAB local unit) and the output signal from the remote unit, using the VCH-314 phase comparator (by Vremya-ch).

All the recorded data were timestamped with 1-s resolution to make it possible to correlate the phase fluctuations with the temperature patterns. The lengths of acquired records were 3–5 days, and they were collected during different time periods.

IV. MODEL OF DCF-INDUCED PHASE FLUCTUATIONS

The testbed described in Section III was used to develop a model of transmission delay correction, based on the recorded DCF temperature variations and simultaneously measured fluctuations of 10 MHz and 1 PPS signals. In order to limit the impact of the noise coming from the temperature sensors, we proposed a modification of (1), which is described in Section IV-A. As the DCF spools display some thermal inertia, the relation between the recorded outer temperature and the resulting delay change should be described in a form of a transfer function (a dynamic model), which is described in Section IV-B.

A. Preprocessing of the Temperature Data

Temperature data from the DCF sensors were read every second to get a reasonable representation of the temperature changes inside the DCF modules. However, as it turned out, temperature values available from the factory-mounted sensors display a fairly large scattering of measured values between consecutive samples (about ± 0.25 K). Thus, the direct usage of the temperature data in (1) produces a very high noise in the calculated phase fluctuations, which scales with the length of the optical fiber used in the DCF module. As the thermal conditions of both DCFs installed at a given node are very similar (see Fig. 2 and related comments in Section II), we proposed to use the averaged readouts from both temperature sensors [ΔT_{AVG_i} in (2)] as a common estimation for forward and backward direction DCF modules in a particular node. Thanks to this modification, the noise of the temperature readouts is reduced, and moreover, its impact scales only with the length difference of two modules installed at each node, i.e., ΔL_{DCF_i} (not as it was before with the length of each DCF). The modified equation takes the form

$$\Delta \tau(t) = \frac{1}{2} A \sum_{i=1}^N (\Delta L_{\text{DCF}_i} \cdot \Delta T_{\text{AVG}_i}(t)). \quad (2)$$

As shown in Fig. 4, the modified formula results in significant (about three times) reduction of the noise level, associated with the temperature measurement. In the calculations, we used the nominal values of the length of the dispersion compensators due to the practical inability to measure all (over 120) DCF modules involved in our measurement setup. Random measurements of such modules indicate a spread of length not exceeding 5% of the rated value.

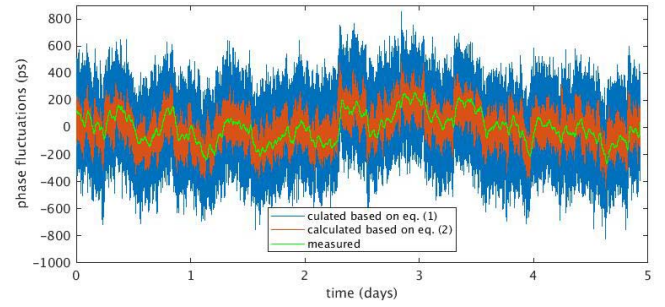


Fig. 4. Phase fluctuations calculated based on (1) and (2) compared to directly measured 10-MHz phase changes.

B. Dynamic Modeling

Equations (1) and (2) adequately describe the temperature-dependent differential delay fluctuations under an assumption that the ambient temperature around the DCF spools changes very slowly, and thus, the temperature inside the spools is uniform and equal to the ambient temperature. We found much better cancelation when the thermal inertia of the DCF spools is taken into account. Thus, we propose a dynamic model based on (2) in the form

$$\Delta \tau(s) = H(s) \frac{1}{2} A \sum_{i=1}^N [\Delta L_{\text{DCF}_i} \cdot \Delta T_{\text{AVG}_i}(s)] \quad (3)$$

where $H(s)$ is the transfer function treated as common (the same) for all spools. To determine the specific form of $H(s)$, we used the MATLAB system identification toolbox [27], fitting the delay fluctuations predicted from (3) to the data recorded directly with the phase comparator. A general form of the transfer function being optimized was

$$H(s) = \frac{\prod (1 + sT_{Zi})}{\prod (1 + sT_{Pi})} e^{-sT_D} \quad (4)$$

where T_{Zi} and T_{Pi} are zeros and poles, respectively, and T_D is a delay (dead time). We examined formulas with up to four poles and two zeros, using data from all three optical routes (described in Section III). Fig. 5 shows the comparison of the effectiveness of considered models with a different number of poles and zeros, and with or without delay, presented in a form of root-mean-square (rms) error between the modeled and directly measured phase fluctuations. As one can see there is no evident advantage of applying higher order dynamic model, which suggests that different factors limit the achievable efficiency of our method (see also comments related to Fig. 6). We also checked that the model optimized based on data from one given route and period works properly applied to different routes and periods, with differences in rms residual error not greater than a several teen percent.

As our goal was to find a relatively simple and universal model, we finally chose the simplest one with just a single pole (no zeros and no delay). The time constant of this pole was taken as an average value obtained from fittings performed on all three routes: $T_P = 1400$ s (which is equivalent to the low-pass cutoff frequency of $110 \mu\text{Hz}$). Optimization of the thermal coefficient of the fiber propagation delay (A) resulted

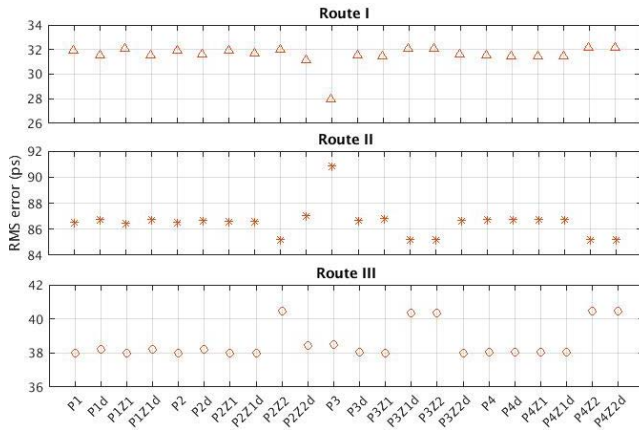


Fig. 5. Comparison of the efficiency of individual models based on rms error. The model name indicates the number of used poles (P), zeros (Z), and using a nonzero delay parameter (d).

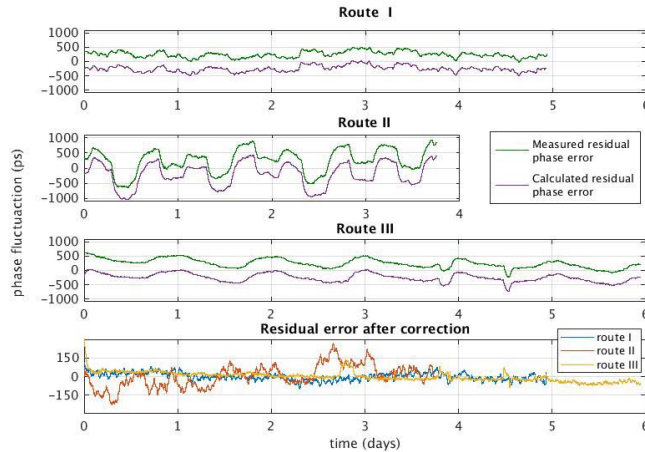


Fig. 6. Phase error of transmitted frequency signals and calculated correction based on DCF temperatures in three different routes. The last graph shows the comparison of absolute error signals, after correction in three routes. The curves on the first three charts are intentionally spaced apart vertically.

in a slight increase of its initially assumed value from 38 to 42 ps/(km·K). This change can be due to the fact that the fibers used for dispersion compensation are significantly different than the regular single-mode fibers in terms of the refractive index profile. Comparison of the described one-pole model with the models obtained for each specific route (i.e., having its time constant individually optimized) shows the difference of the rms error not greater than 3 ps, which confirms its universal character.

Fig. 6 shows the time plots of the 10-MHz phase instability, measured directly and calculated on the basis of the described model, for all three routes. The second (II) route had the largest rms value of phase changes (390 ps), and the rms fluctuations of the corrected phase are reduced down to 90 ps. The lowest rms value of the corrected phase was observed in route (I)—30 ps versus 110 ps before correction. Commenting on noticeable imperfectness of the phase correction, two groups of reasons should be pointed out. The first is related to our modeling and comprises the inaccuracy

of temperature sensors and accurate data concerning DCF lengths. In this aspect, the accuracy of our method could be basically improved. The second group, however, includes the effects ignored in our model, as the thermal behavior of various optical modules/devices presents at each network node, or possibly unequal length or nonidentical temperature distribution along the forward and backward fiber paths.

V. FREQUENCY STABILITY AND ACCURACY AFTER CORRECTION

The previously described model of the impact of DCF modules on the fluctuations of differential delay can be used for improving the phase stability. In the case of comparing two distant atomic clocks, the calculated impact of DCFs can be subtracted from actually measured mutual instability of the clocks, and this way, the negative influence of the link can be significantly reduced.

In our experiment, the results obtained after subtracting the calculated impact of DCFs from the raw phase fluctuation's data, measured with the phase comparator, illustrate the improvement in the link stability and accuracy.

Fig. 7 shows the modified Allan deviation (MDEV) of 10-MHz signal before and after correction for three different analyzed routes. As it can be seen, significant improvement in stability is obtained starting from about 500 s of averaging time. In the long-term averaging, the signal stability is improved from five to ten times.

For medium-term averaging (50 to 500 s), a small increase of the MDEV is, however, noticeable probably due to imperfect dynamic modeling of the influence of ambient temperature on DCF delay.

In turn, for short averaging, the stability of the frequency transfer is unaffected by our method. First of all, it is due to a different (not thermally induced) origin of the short-term phase noise. In addition, it may be noticed that the modeling of the thermal inertia of DCF modules acts as low-pass filtering of the data from the temperature sensors, which fortunately suppresses the noise of the sensors to the level not deteriorating short-term stability.

In order to present the accuracy of the frequency transfer, the averaged frequency offsets for 1 h, 10^4 s, and one day averaging were analyzed. Table I summarizes the maximum and rms frequency offsets for different averaging periods. In all situations, the proposed frequency correction caused a visible improvement in frequency transfer accuracy. In most cases, the reduction of the frequency offset was in the range of two to five times. In addition, in Fig. 8, we show a time evolution of 1-h averaged frequency offsets in entire measurement periods. The offsets after correction seem to be randomly distributed, and show no evident correlation with the situation before correction.

VI. IMPACT OF REDUCED DCF TEMPERATURE POLLING PERIOD

From the point of view of the operation of wide area network management, the polling of a very large number of devices with a 1-s repetition via SNMP is cumbersome.

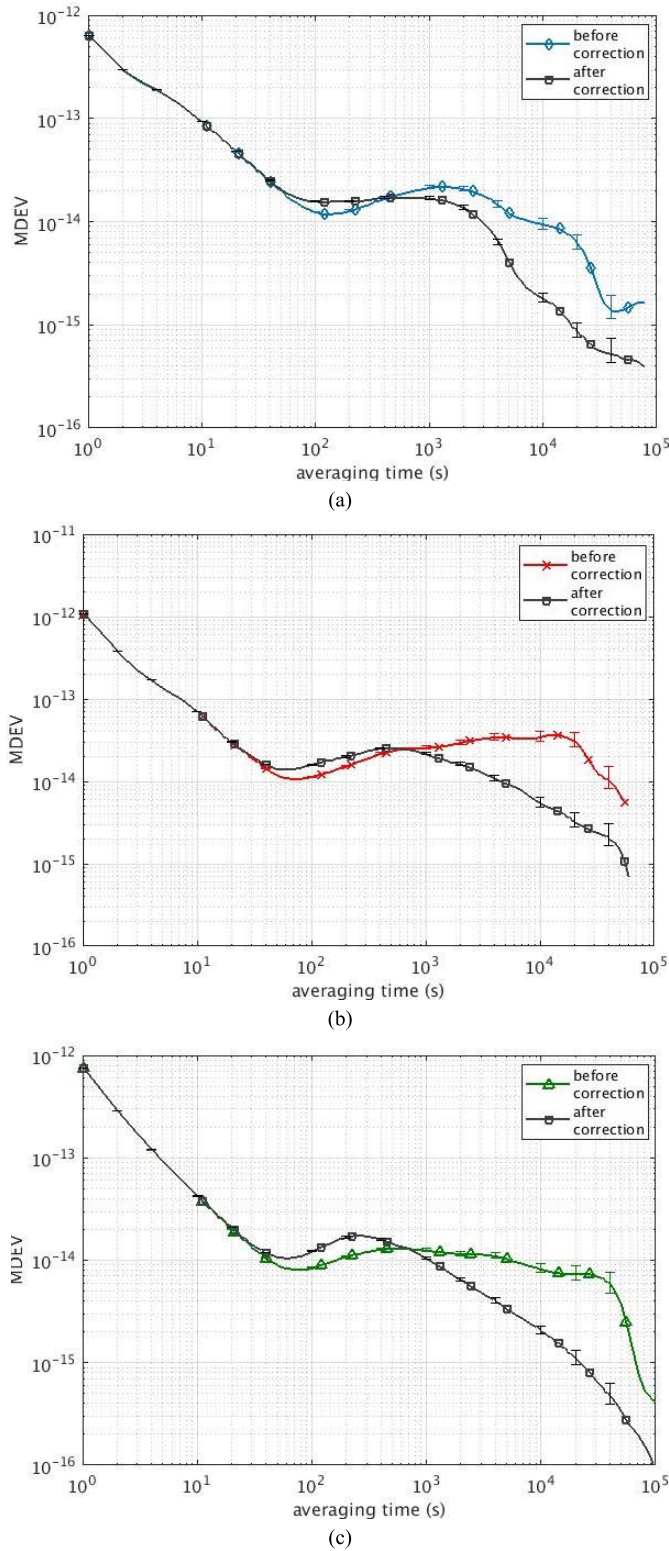


Fig. 7. MDEV for 10-MHz phase changes before and after correction in routes (a) I, (b) II, and (c) III.

Such frequent sampling of devices causes both a heavy load on the management server and the necessity of storing a large amount of data.

Therefore, in the following, we present an analysis of the effectiveness of the proposed algorithm in the case of the

TABLE I
FREQUENCY OFFSET IN VARIOUS DWDM ROUTES, BEFORE AND AFTER CORRECTION WITH DIFFERENT AVERAGING TIMES

Averaging time		I	II	III
1 h	Before max / RMS	$6.1^{-14} / 2.2^{-14}$	$1.8^{-13} / 6.0^{-14}$	$1.2^{-13} / 1.8^{-14}$
	After max / RMS	$3.0^{-14} / 1.1^{-14}$	$4.2^{-14} / 1.6^{-14}$	$7.6^{-14} / 8.9^{-15}$
10 ⁴ s	Before max / RMS	$3.4^{-14} / 1.4^{-14}$	$1.0^{-13} / 4.8^{-14}$	$4.6^{-14} / 1.3^{-14}$
	After max / RMS	$8.0^{-15} / 2.9^{-15}$	$2.0^{-14} / 1.1^{-14}$	$2.3^{-14} / 4.6^{-15}$
1 day	Before max / RMS	$3.5^{-15} / 2.3^{-15}$	$4.9^{-15} / 3.0^{-15}$	$2.7^{-15} / 1.5^{-15}$
	After max / RMS	$2.8^{-16} / 1.5^{-16}$	$2.0^{-15} / 1.7^{-15}$	$5.3^{-16} / 3.5^{-16}$

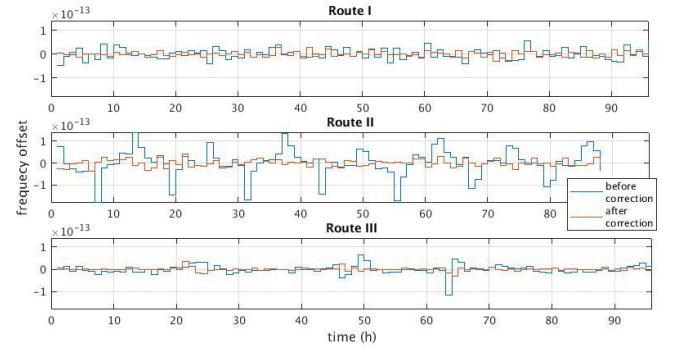


Fig. 8. Frequency offset (1-h averaged) before and after correction calculated for all three routes.

reduced sampling rate of the temperature data collection. For this purpose, we used DCFs' temperatures for the (I) route, where the previously collected 1-s spaced data were decimated (i.e., all but one readouts were ignored for a given period) to simulate the reduced rate of queries. Then, the decimated data obtained in this way were subjected to interpolation in order to obtain strings of input data with 1-s resolution, consistent with the previous calculations. We repeated the calculations with the sampling period increased to 10, 30, 60, 120, 300, and 600 s. Fig. 9 shows the stability analysis of the corrected signal for two selected sampling rates compared with nondecimated one. Changing the repetition period of DCF data collection in the range from 10 to 60 s leads to a gradual deterioration of stability in the range of 50–1000 s averaging time, but this deterioration is not greater than by a factor of 1.5. The charts nearly overlap for long averaging times. After decreasing the sampling rate over 120 s, the aliasing effect worsens slightly the stability as well for long averaging times (for sampling 300 s, long-term stability is decreased by a factor of 1.3).

From the point of view of a compromise between the requirements of a network management system and the efficiency of compensation, it seems that the reasonable sampling rate of DCF temperatures is 60 s.

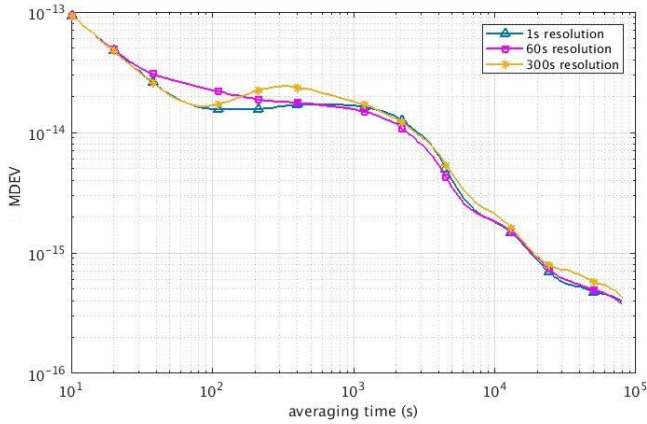


Fig. 9. Frequency stability after correction shown for different sampling rates of the temperature variations in the DCFs.

Due to the fact that the correction of the transmitted signal takes place in the postprocessing process, it is necessary to remember about the synchronization of servers polling the DCF temperatures and the devices measuring the fluctuations of the transmitted signal (phase comparator). In practice, it is sufficient to maintain the 1-s level of synchronization. This requirement is not so stringent and can be accomplished using a common network time protocol server.

VII. CONCLUSION

It has been demonstrated that the DCFs have a significant influence on the frequency transfer stability in DWDM networks. It was also shown that this impact can be significantly reduced by postprocessing of the raw stability data, by taking into account the model of the DCFs' influence based on their temperature information. Experiments performed showed that the presented method allows improving significantly the long-term stability of the frequency signal.

The proposed method has been verified on various routes to confirm the universality of the applied method. However, it seems reasonable to verify model parameters (A and T_p) when the target network is based on different equipment families, especially when a different type of DCF spools is used.

Although this paper is focused on the frequency transfer during these experiments, we simultaneously carried out the time transfer (1 PPS pulses) and monitored its stability with high-resolution time interval counter. After applying the same phase corrections as for frequency transfer, we obtained a visible improvement of the time transfer stability. Fig. 10 shows an example of the correction result in route I, presented in the form of a time deviation (TDEV) plot. As for the frequency signal, the proposed method significantly improves the long-term stability; TDEV is reduced to below 20 ps.

The proposed method of the stability improvement, demonstrated in a form of postprocessing of the collected phase data, might be also implemented for a real-time correction of the output frequency (and time) signal outgoing the remote terminal. For this purpose, the online calculated phase corrections should be provided to a hardware phase shifter added at the remote terminal. According to our experiments (see Fig. 6),

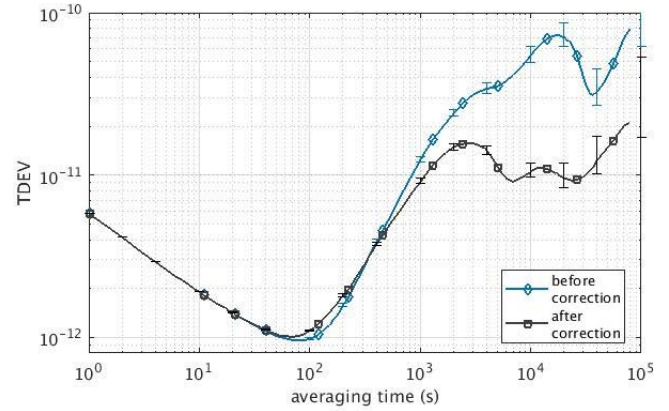


Fig. 10. TDEV for 1 PPS phase changes before and after correction in route I.

the relatively low tuning range of such shifter, as ± 1 ns, seems to be sufficient to compensate the impact of DCFs.

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